

Thermo-VLA: A Thermal-Aware Vision-Language-Action Model with Gated Residual Modulation for Robotic Manipulation (Supplementary Materials)

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I. SYMBOL NOTATION

For clarity and consistency, Table S1 summarises the main notation used throughout this work, including observations, visual and attribute representations, model components, gating variables, robot states and actions, and flow-matching variables.

S1: **Notation.**

Symbol	Description
x	Original RGB visual representation
a	Attribute observation
l	Language instruction or condition
$f(\cdot)$	Function that outputs the additive residual
y	Attribute-modulated RGB representation
x_r	RGB observations
x_r^i	RGB observation from camera i
x_t	Thermal observation
x_{cold}	Cold attribute view
x_{hot}	Hot attribute view
x_{col}^j	Colour attribute view for colour j
Z_{rgb}^i	RGB tokens from camera i
$Z_{\text{rgb}}^{\text{head}}$	Head-camera RGB tokens
$Z_{\text{rgb}}^{\text{left}}$	Left wrist-camera RGB tokens
$Z_{\text{rgb}}^{\text{right}}$	Right wrist-camera RGB tokens
Z_{cold}	Cold attribute tokens
Z_{hot}	Hot attribute tokens
$Z_{\text{rgb}}^{\text{cold}}$	Cold-conditioned RGB representation
$Z_{\text{rgb}}^{\text{hot}}$	Hot-conditioned RGB representation
Z_{col}^j	Colour attribute tokens for colour j
Z_{rgb}^j	Colour-conditioned RGB representation for colour j
Z_{text}	Joint textual representation of language and robot state
Z_T	Temperature-conditioned multimodal prefix
E_r	RGB visual encoder
E_t	Shared thermal visual encoder
E_{col}	Shared colour visual encoder

Symbol	Description
P_r	Visual projector
P_{col}	Colour visual projector
O_c	Cross-attention output of the cold branch
O_h	Cross-attention output of the hot branch
u_t	Gating feature
α, β	Dynamic weights for cold and hot attributes
α', β', γ	Dynamic weights for red, green, and blue attributes
LN	Layer normalization
MLP	Gated multilayer perceptron
s_t	Robot state at time step t
a_t	Robot action at time step t
A_t	Action chunk starting at time step t
H	Action horizon
d_a	Single-arm action dimension
π_θ	Parameterised action policy
θ	Learnable policy parameters
i	Camera index, $i \in \{\text{head, left, right}\}$
j	Colour index, $j \in \{\text{red, green, blue}\}$
B	Batch size
N	Number of visual tokens
D	Token dimensionality
b	Background reference thermal response
μ	Fixed background grayscale intensity
k	Background-calibrated thermal observation
I_{max}	Upper bound of the grayscale range
ϵ	Gaussian noise, $\epsilon \sim \mathcal{N}(0, I)$
I	Identity matrix
τ	Flow-matching time step
x_τ	Interpolated action variable at flow time τ



Fig. S1. **Equipment:** Unitree G1 EDU platform, sensing setup, and manipulation objects across 7 tasks.

II. HARDWARE DETAILS

Experiments are conducted on a Unitree G1 EDU humanoid robot equipped with Unitree Dex3-1 three-finger dexterous hands, with only the two arms and hands being controlled. Visual observations are captured by a head-mounted Intel RealSense D435i RGB camera (640×480 , 30 Hz), two wrist-mounted Intel RealSense D405 cameras, and a Tiny1-C thermal camera (256×192 , 3.2 mm, 25 Hz) mounted to the right of the D435i. Model training is performed on an NVIDIA A800 80GB GPU, while policy inference runs on an NVIDIA GeForce RTX 5090 48GB GPU. Both systems use Ubuntu 22.04. As shown in Fig. S1, the RGB and thermal cameras are rigidly mounted on the robot head, while the two wrist cameras are installed on the left and right wrists following Unitree’s official mounting configuration. The figure also presents the complete set of manipulation objects used throughout our experiments, including objects used for task data collection and additional objects introduced for unseen-object generalisation evaluation. Several objects are fabricated via 3D printing using publicly available models from MakerWorld.

III. TRAINING DETAILS

A. Training Budget Selection

Fig. S2 shows the training loss curves for the six tasks. Across all tasks, the loss decreases rapidly during the early stage of training and gradually converges after approximately 20K–30K iterations. By 40K iterations, the loss curves have reached consistently low and stable levels, while further training yields only marginal reductions with continued stochastic fluctuations. We therefore use 40K iterations as the training budget and select the corresponding checkpoint for evaluation. For reproducibility, the random seed is fixed to 1000 for all experiments.

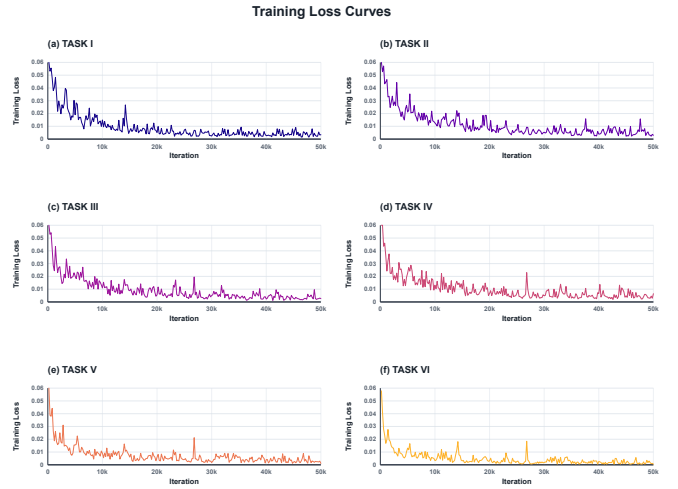


Fig. S2. **Training loss curves.** All tasks converge to stable loss values within the 40K-step fine-tuning budget used in our experiments.

IV. DATASET

A. XR Teleoperation

Datasets are collected following the XR teleoperation pipeline provided in Unitree Robotics’ official XR teleoperate repository. We use a PICO XR system and develop a semi-automated collection procedure, where each recorded episode includes an automatic transition from the predefined initial pose to the operator’s current pose, teleoperated task execution, and an automatic return to the initial pose after task completion. In addition to the original robot states and RGB observations, thermal observations are synchronously recorded and integrated into an extended LeRobot v3 data format. This unified multimodal dataset is used for all VLA training and evaluation in this work.

B. Thermal Objects Elimination

In Task I and Task III, the target objects should not provide task-relevant temperature cues, while the thermal radiation from the robot hands should still be preserved. To reuse the thermal data collected in Task II and Task IV, we remove the thermal responses of target objects and retain only the robot-hand heat signatures. We explore two segmentation approaches, RobotSeg and Cutie.

RobotSeg. RobotSeg [1] is a robot-specific segmentation model built upon SAM 2, combining structure-enhanced temporal memory with an automatic robot prompt generator. We use it to segment the robot regions in thermal sequences, retain the arm and hand signals, and replace the remaining regions with the background mean value. While RobotSeg can reliably localise a visible arm, it may fail when the hand becomes occluded, as shown in Fig. S3.

Cutie. Cutie [2] is a video object segmentation model based on object-level memory and query-based feature interaction. We manually annotate the temperature-bearing target objects in a few key frames and propagate their masks throughout the sequence using Cutie. The segmented target regions are then removed, followed by thresholding

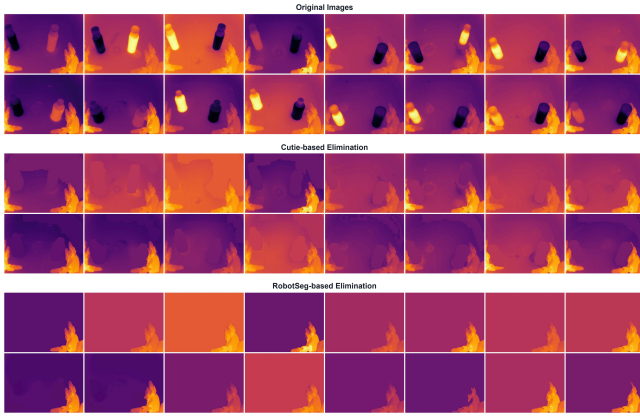


Fig. S3. **Thermal-object elimination for constructing RGB-only observations.** RobotSeg directly extracts the robot regions but may lose the hand under occlusion, whereas Cutie tracks and removes temperature-bearing target objects more consistently while preserving the thermal responses of the robot hands.

to suppress residual low-temperature responses. Compared with RobotSeg, Cutie provides more stable segmentation and produces cleaner thermal observations consistent with the RGB-only settings of Task I and Task III. Qualitative results are shown in Fig. S3.

C. Dataset Details

Our dataset is designed to evaluate both conventional RGB manipulation capability and progressively more complex thermal reasoning. Tasks I and III are RGB-only tasks based on spatial position and object identity, respectively, while Tasks II, IV, V, and VI progressively increase the thermal reasoning difficulty: from binary temperature-based selection, to target selection among multiple objects, fine-grained temperature discrimination in a drink-service scenario, and abnormality reasoning in a laboratory scenario. Tasks V and VI further move beyond basic pick-and-place settings toward more practical and long-horizon manipulation.

Tasks VII and VIII further examine whether the proposed residual formulation can generalise beyond thermal attributes by using colour as the task-relevant cue. Task VII is conducted in simulation because existing robotic simulators do not provide sufficiently realistic thermal observations, while Task VIII provides the corresponding real-world evaluation. Table S2 summarises the language instructions and number of collected episodes for all tasks.

D. Training Parameters

All $\pi_{0.5}$ -based models are initialised from the same pre-trained $\pi_{0.5}$ checkpoint and fine-tuned under identical settings with AdamW for 40K steps (batch size 8, learning rate 2.5×10^{-5} , 100-step warm-up, and cosine decay). Other baselines follow their corresponding architectures and training configurations as specified in their original papers, while using the same task data and evaluation protocol; further implementation details are provided in Table S3.

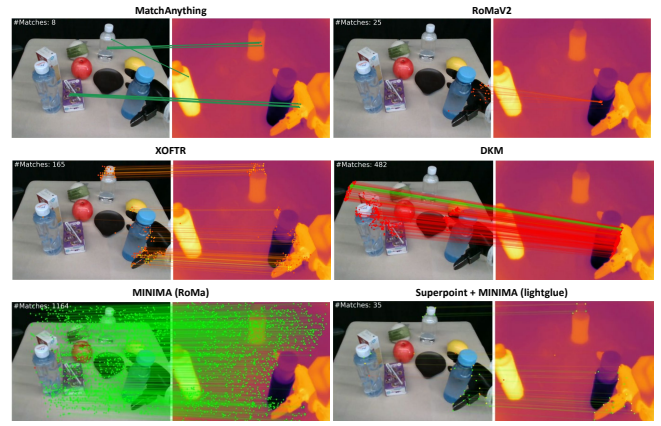


Fig. S4. **Comparison of RGB-thermal cross-modal matching methods.** We compare MatchAnything, RoMaV2, XoFTR, DKM, MINIMA (RoMa), and SuperPoint + MINIMA (LightGlue). Among the compared methods, SuperPoint + MINIMA (LightGlue) provides the most stable correspondences in our setting and is therefore used in our explicit-alignment extension.

V. THERMO-VLA WITH RGB-THERMAL ALIGNMENT

A. Motivation

The current implementation of Thermo-VLA uses rigidly mounted RGB and thermal cameras, such that their relative pose remains fixed during both data collection and inference. Under this setting, it is first important to examine whether explicit RGB-thermal matching is necessary, or whether the learned multimodal representation can already tolerate the fixed spatial offset between the two sensing streams. Meanwhile, in more general robotic deployments, the thermal camera may be placed at different positions or viewpoints, leading to larger cross-modal misalignment. We therefore investigate explicit cross-modal matching both to assess its necessity under the default fixed-viewpoint setting and to explore a potential extension that relaxes Thermo-VLA’s dependence on a fixed RGB-thermal configuration without modifying its core architecture.

B. Matching Algorithm

To identify a suitable RGB-thermal matching method, we compare MatchAnything [3], RoMaV2 [4], XoFTR [5], DKM [6], MINIMA (RoMa) [7], [8], and SuperPoint + MINIMA (LightGlue) [9], [7], [10], as shown in Fig. S4. The MINIMA-based methods yield more reliable cross-modal correspondences, with SuperPoint + MINIMA (LightGlue) providing the most stable results in our setting. We therefore adopt this configuration.

SuperPoint [9] extracts sparse local features from RGB and thermal images, while the MINIMA-adapted LightGlue [10], [7] establishes correspondences for spatial pre-alignment before thermal preprocessing. Under the same tasks and viewpoints as the main experiments, explicit matching achieves comparable success rates to standard Thermo-VLA but introduces additional inference latency. This suggests that explicit alignment is unnecessary under the default fixed-viewpoint setting, while providing a practical extension when larger RGB-thermal misalignment occurs.

TABLE S2

DATASET DETAILS. LANGUAGE INSTRUCTIONS AND NUMBER OF COLLECTED EPISODES FOR EACH TASK. FOR TASKS I AND II, $x + y$ DENOTES EPISODES COLLECTED WITH THE TWO BOTTLE TYPES, WHERE x AND y CORRESPOND TO THE SHORTER AND BLUE BOTTLES, RESPECTIVELY.

Task Name	Language Instructions	Episode
Task 1: Position-guided bottle selection		
	Pick up the bottle on the left and place it on the black circular coaster.	39 + 42
	Pick up the bottle on the right and place it on the black circular coaster.	46 + 29
Task 2: Temperature-guided bottle selection		
	Pick up the cold bottle and place it on the black circular coaster.	152 + 151
Task 3: Identity-guided object selection		
	Pick up the milk and place it on the white circular coaster.	48
	Pick up the milk and place it on the yellow circular coaster.	56
	Pick up the coke and place it on the white circular coaster.	71
	Pick up the coke and place it on the yellow circular coaster.	61
Task 4: Temperature-guided object selection		
	Pick up the hot object and place it on the yellow circular coaster.	89
	Pick up the hot object and place it on the white circular coaster.	84
	Pick up the cold object and place it on the yellow circular coaster.	83
	Pick up the cold object and place it on the white circular coaster.	84
Task 5: Temperature-aware coffee service		
	Pick up the sugar container and add sugar to the coffee with less ice.	87
	Pick up the sugar container and add sugar to the coffee with extra ice.	86
Task 6: Abnormal-animal cage removal		
	Select the cage with the dead mouse and place it in the hazard-marked box.	99
Task 7: Colour-guided cube selection (Simulation)		
	Put the red cube on the plate	50
	Put the green cube on the plate	50
	Put the blue cube on the plate	50
Task 8: Colour-guided cup selection		
	Select the red cup and place it on the black coaster	73
	Select the blue cup and place it on the black coaster	51

TABLE S3

IMPLEMENTATION CONFIGURATION OF THE THERMORES ENCODER. THE TABLE SUMMARISES THE BACKBONE AND INPUT SETTINGS, THERMAL ATTRIBUTE DECOMPOSITION, VISUAL ENCODING AND GATING, AND ACTION-GENERATION CONFIGURATION.

Component	Configuration	Value
Backbone and Input		
Backbone	Vision–language backbone / action expert	2B / 300M parameters
Input geometry	Image resolution / visual tokens / token dimension	224×224 / 256 / 2048
Sequence configuration	Maximum textual length / action horizon	200 / 50
Thermal Attribute Separation		
Background estimation	Spatial sampling regions	$3 \times 64 \times 64$ upper-boundary regions
Background alignment	Reference grayscale intensity	$\mu = 80$ on $[0, 255]$
Outlier filtering	Robust threshold / relative threshold	12 / 2.0
Visual Encoding and Gating		
Visual encoders	RGB encoder / thermal encoder	Independent / shared across 2 branches
Visual projection	Projector across visual streams	Shared P_r
Cross-attention	Textual query / thermal key–value	$Q = Z_{\text{text}}, K = V \in \{Z_{\text{cold}}, Z_{\text{hot}}\}$
Attention configuration	Number of heads / head dimension	8 / 256
Gate network	MLP dimensions	$10240 \rightarrow 512 \rightarrow 256 \rightarrow 2$
Gate activation	Hidden activation / dropout	GELU / 0
Gate initialization	Output weight / bias	$\mathcal{N}(0, 0.02)$ / 0
Output and Action Generation		
Dynamic gating	Cold / hot branch weights	$[\alpha, \beta] = \text{softmax}(\text{MLP}(\text{LN}(u_t)))$
Action generation	Action embedding dimension / integration steps	1024 / 10

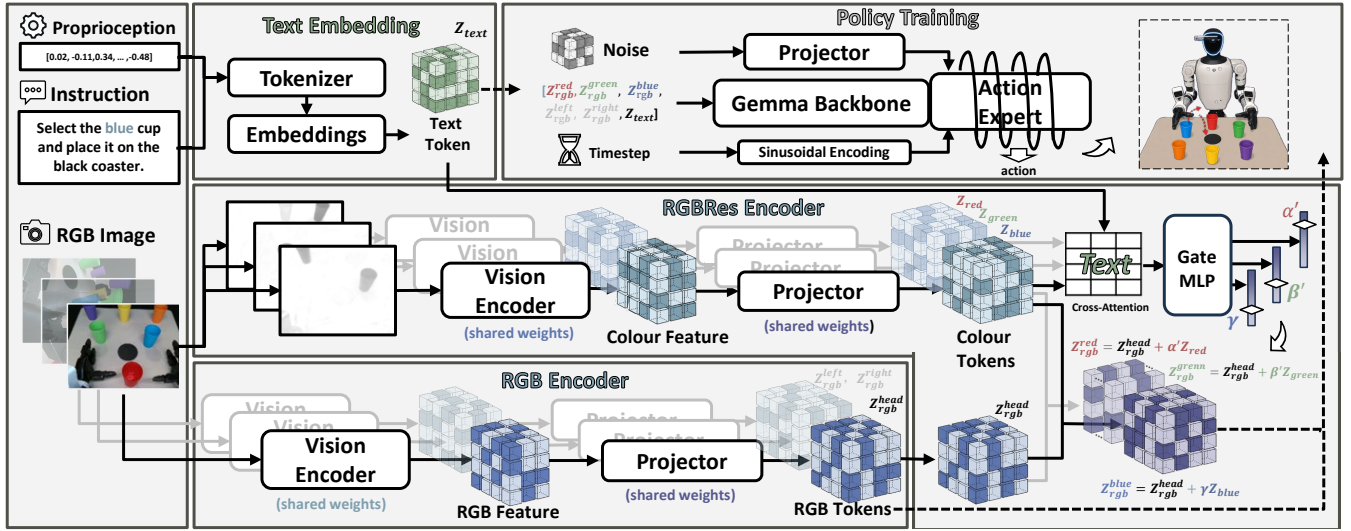


Fig. S5. **Overview of $\pi_{0.5}$ with the RGBRes Encoder.** Attribute-specific RGB views are encoded as residual representations and injected into the original RGB representation through language-conditioned gating, extending the residual formulation beyond thermal attributes.

VI. OVERVIEW OF $\pi_{0.5}$ WITH RGBRES ENCODER

As shown in Fig.S5, we extend $\pi_{0.5}$ with an additional RGBRes Encoder to introduce attribute-conditioned residual modulation into the RGB visual representation.

VII. LOW-RANK ANALYSIS OF RESIDUAL MODULATION

A. Preliminaries

We first recall the classical Eckart–Young–Mirsky theorem for optimal low-rank approximation [11], [12].

Theorem 1 (Eckart–Young–Mirsky). Let $M \in \mathbb{R}^{m \times d}$ be a matrix of rank r , with singular value decomposition

$$M = U \Sigma V^\top = \sum_{j=1}^r \sigma_j u_j v_j^\top,$$

where

$$\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0.$$

For any integer $1 \leq k < r$, define the rank- k truncated singular value decomposition as

$$M_k = \sum_{j=1}^k \sigma_j u_j v_j^\top.$$

Then M_k is a best rank- k approximation of M under the Frobenius norm, i.e.,

$$M_k \in \arg \min_{\text{rank}(A) \leq k} \|M - A\|_F,$$

and the minimum approximation error is

$$\min_{\text{rank}(A) \leq k} \|M - A\|_F = \|M - M_k\|_F = \left(\sum_{j=k+1}^r \sigma_j^2 \right)^{1/2}.$$

Equivalently,

$$\min_{\text{rank}(A) \leq k} \|M - A\|_F^2 = \sum_{j=k+1}^r \sigma_j^2.$$

If $\sigma_k > \sigma_{k+1}$, the optimal rank- k approximation is unique.

B. Low-Dimensional Residual Modulation

We now apply Theorem 1 to the residual formulation used in our framework. Let

$$y = x + \delta, \quad \delta = f(a; l),$$

where $x \in \mathbb{R}^D$ denotes the original RGB representation, a denotes the corresponding attribute representation, l denotes the language condition, and $\delta \in \mathbb{R}^D$ denotes the resulting attribute-conditioned residual.

For spectral analysis, each D -dimensional residual token is treated as an individual sample, and residuals across observations and attribute branches are pooled together. Denoting the resulting n residual samples as $\{\delta_i\}_{i=1}^n$, we first remove their common offset:

$$\bar{\delta} = \frac{1}{n} \sum_{i=1}^n \delta_i,$$

and construct the centred residual matrix

$$\tilde{D}_\delta = \begin{bmatrix} (\delta_1 - \bar{\delta})^\top \\ \vdots \\ (\delta_n - \bar{\delta})^\top \end{bmatrix} \in \mathbb{R}^{n \times D}.$$

Let

$$\tilde{D}_\delta = U \Sigma V^\top$$

be its singular value decomposition, and let $V_k = [v_1, \dots, v_k] \in \mathbb{R}^{D \times k}$ contain the first k right singular vectors.

We define the cumulative spectral energy retained by the top- k components as

$$\eta_k := \frac{\sum_{j=1}^k \sigma_j^2}{\sum_{j=1}^r \sigma_j^2}.$$

According to Theorem 1, the optimal rank- k approximation $\tilde{D}_\delta^{(k)}$ satisfies

$$\frac{\|\tilde{D}_\delta - \tilde{D}_\delta^{(k)}\|_F^2}{\|\tilde{D}_\delta\|_F^2} = 1 - \eta_k.$$

Hence, if η_k approaches 1 for $k \ll D$, the centred residual variations can be accurately represented in the k -dimensional subspace spanned by V_k . Specifically, defining

$$z_i = V_k^\top (\delta_i - \bar{\delta}) \in \mathbb{R}^k,$$

the corresponding low-rank reconstruction is

$$\delta_i \approx \bar{\delta} + V_k z_i.$$

Substituting this result into the original additive formulation gives

$$y_i \approx x_i + \bar{\delta} + V_k z_i, \quad k \ll D.$$

Corollary 1. If the centred residual matrix satisfies $\eta_k \approx 1$ for some $k \ll D$, then the residual variation admits an approximately low-dimensional affine representation. Accordingly, the additive formulation can be expressed as a shared residual offset $\bar{\delta}$ together with a low-dimensional sample-dependent component $V_k z_i$ added to the original RGB representation.

C. Empirical Verification

As shown in Table S4, the residual representations broadly satisfy Corollary 1. Here, PCA is performed over the pooled token-level residual features described above. For the RGB simulation tasks, the top 8 principal components retain 95.7% and 94.0% of the centred residual variation in Task VII and Task VII^{Table}, respectively, indicating that the unseen tabletop introduces little additional variation. The real-world thermal tasks are less concentrated, with the top 32 components retaining 83.1%, 84.4%, and 89.6% in Tasks II, IV, and VI. Given the token dimension $D = 2048$, most residual variation is still captured in a substantially lower-dimensional subspace. Equivalently, the empirical spectra support the approximation

$$\delta_i \approx \bar{\delta} + V_k z_i, \quad y_i \approx x_i + \bar{\delta} + V_k z_i,$$

with $k \ll D$.

We additionally measure the average pairwise cosine similarity between the residual branches. The similarity ranges from 0.942 to 0.956 for the thermal tasks and reaches 0.999 for the RGB tasks, showing strong directional alignment among attribute residuals. Together with the centred PCA results, these observations support interpreting the learned residuals as an approximately low-dimensional attribute-conditioned additive modulation of the original RGB representation.

TABLE S4

GEOMETRIC ANALYSIS OF ATTRIBUTE-CONDITIONED RESIDUAL REPRESENTATIONS. η_k DENOTES THE CUMULATIVE CENTRED PCA ENERGY RETAINED BY THE TOP- k PRINCIPAL COMPONENTS.

Task	Avg. Cosine	η_4	η_8	η_{16}	η_{32}
Task II	0.942	54.2	66.1	74.5	83.1
Task IV	0.947	58.3	69.3	76.5	84.4
Task VI	0.956	69.6	79.6	84.5	89.6
Task VII	0.999	88.1	95.7	98.4	99.5
Task VII ^{Table}	0.999	84.8	94.0	97.6	99.1

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